

Chapter 61 — Urologic Considerations in Pediatric Renal Transplantation — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Safe storage:** detrusor pressure <40 cm H₂O at age-expected capacity.
- **Augment:** capacity <75% expected, pressure >40, CIC (clean intermittent catheterisation) every ≤3 h, despite maximal therapy.
- **Never use the allograft to test the bladder.**
- **Preferred anastomosis:** extravesical non-refluxing ureteroneocystostomy.
- **Salvage for graft-ureter stenosis:** ureteroureterostomy to the native ureter.
- **Commonest urological complication:** ureteric stenosis (6%).
- **Obstruction:** >half within **100 days**; usually **distal** at the reimplant.
- **Creatinine rise + hydronephrosis:** catheter first; obstruction and rejection may coexist.
- **Transplant VUR** (vesicoureteral reflux) + recurrent febrile UTI (urinary tract infection) → reimplant; asymptomatic → observe.
- **Default:** keep native kidneys; remove only for a named indication.

INTRODUCTION AND PRETRANSPLANT ASSESSMENT

- **Five goals:** unimpeded drainage · low-pressure capacious storage · volitional emptying · no infection · fewest operations.
- **Commonest age at transplant:** 13–17 years.
- **Commonest congenital cause:** hypoplasia, aplasia or dysplasia, then obstructive uropathy; **commonest acquired:** FSGS (focal segmental glomerulosclerosis).
- **Young boys** → PUVs (posterior urethral valves); **adults** → hypertension, diabetes.
- **Screen all:** history + renal ultrasound + uroflow with PVR (postvoid residual).
- **Routine VCUG (voiding cystourethrogram) not needed;** do it if neuropathic bladder, obstruction, VUR, previous surgery, febrile UTI.
- **Urodynamics:** neuropathic bladder, PUVs or large ureterocele, refractory dysfunction, high-grade hydronephrosis, recurrent UTI.
- **Anuric, no urological history** → no further testing.
- **Algorithm:** **small capacity** <30% → anticholinergic and reassess.

PRETRANSPLANT PREPARATION

Bladder:

- **Commonest early ESKD (end-stage kidney disease) bladder:** low capacity, overactivity, poor compliance.
- **Most dangerous:** overactivity and poor compliance (pressure reaches the kidney without reflux).
- **Valve bladder trajectory** → **myogenic failure.**
- **First line:** antimuscarinic (oxybutynin, titrate up); **augmentation second line.**
- **Better tolerated:** solifenacin, darifenacin, **mirabegron** (β₃-agonist), botulinum toxin.
- **α-blocker** → aids emptying.
- **Volitional emptying never assumed;** prove it with repeated PVR and uroflow.
- **CIC adherence** = test of transplant adherence.
- **Cycling target:** expected capacity for age; confirm by urodynamics.

Augmentation:

- **No evidence augmentation raises transplant complications.**
- **Ileum or sigmoid** = commonest segments; malignancy risk → lifelong surveillance.
- **Gastric:** haematuria–dysuria syndrome, worst in anuric patients; rarely used.
- **Ureterocystoplasty:** always explore first; no mucus, stones or metabolic disturbance.

- **PUV bladder** → delay augmentation; **neuropathic, borderline** → augment earlier.
- Stomas sited **medial** to the transplant incision.
- **Gibson incision, retroperitoneal** = usual graft site; <15 kg → aorta and IVC (inferior vena cava), possibly transperitoneal.

Timing:

- **Reconstruct before transplant; 6–12 weeks** to heal after enterocystoplasty.
- **Exception:** transplant in the **first 2 years** → defer continent diversion.
- **No concurrent procedures in infants;** anticoagulation raises bleeding.
- **Metabolic problems** = not an absolute contraindication to diversion.

Dialysis:

- **On PD (peritoneal dialysis):** intraperitoneal surgery → temporary haemodialysis.
- PD + small kidneys → **retroperitoneal (prone bilateral);** PD restarts immediately.
- PD + large kidneys → posterior open or laparoscopic morcellation.
- High risk → **embolisation.**

Native nephrectomy indications:

- **Refractory hypertension · nephrotic syndrome with malnutrition · massive polyuria · recurrent upper-tract UTI · large stone burden · malignancy risk · very large kidneys · massive VUR (relative).**
- **Keep otherwise:** water excretion, erythropoietin, vitamin D.
- **Combined with transplant:** avoid; best = single ipsilateral nephrectomy through the transplant incision.

Ureter:

- **Normal ureter** → keep; transect high (salvage ureteroureterostomy, stoma).
- **Dilated, refluxing or obstructed** → remove totally.
- **Circumcision** in boys with valves → fewer UTIs.
- **Ileal loop** → augmentation patch, or discard after native reimplant.

TRANSPLANTATION

- **Extravesical:** detrusorotomy 3–4 cm; spatulate 4–5 mm; mucosa-to-mucosa.
- **Barry technique:** 4-cm tunnel between parallel incisions.
- **Ureteroureterostomy:** normal native ureter, small bladder, short donor ureter, hostile pelvis.
- **Routine stent:** no evidence; stent if reimplant difficult or ureter devascularised.

- **Double-J out at 4–6 weeks; early removal at 5–7 days** → fewer infections.

COMPLICATIONS

- **Thrombosis** = a leading cause of **early graft loss** in children.
- **Pre-existing uropathy** → more urological complications.
- **Rates**: stenosis 6% (obstruction up to 8%), leak 3% (<5%), stones 2%, significant reflux 2%.
- **Leak**: early; **fluid creatinine** confirms; check the catheter first.
- Localise the leak with **MAG3** (mercaptoacetyl triglycine scan) or cystogram.
- **Largest catheter** the urethra takes = prevention.
- **Long ureteric necrosis** → native ureter (ipsilateral or contralateral), psoas hitch, bladder flap.
- **Endoscopic injection for graft VUR**: about one-third resolve; higher obstruction risk.
- **Reimplant for graft VUR**: intravesical preferred; stent.
- **≥2 UTIs** → lower graft GFR (glomerular filtration rate) at 4 years.
- **Mild hydronephrosis + signs of rejection** → biopsy first.
- **Bladder dysfunction test**: catheter **1–2 weeks** → creatinine falls → bladder is the cause.
- **PUVs** → lower obstruction-free survival.
- **Focal stenosis** → balloon dilatation + stent **4–6 weeks**.
- **Lymphocele, PTLD** (post-transplant lymphoproliferative disease) = extrinsic causes; **graft mass = PTLD** until proven otherwise.

- **Stones**: treat if **>1 cm and growing**, obstructing or infected; choose single-intervention success.
- **Donor stones**: bench flexible ureteroscopy; donor-gifted stones observed (adult data).

CLASSIC TRAPS

- **Trap**: test the bladder after transplant — **Actually**: never use the graft to test the bladder.
- **Trap**: no reflux means a safe bladder — **Actually**: high pressure harms the kidney without reflux.
- **Trap**: augmentation raises transplant risk — **Actually**: no compelling evidence.
- **Trap**: gastric augment for an acidotic child — **Actually**: haematuria-dysuria syndrome; abandoned.
- **Trap**: augment the PUV bladder early — **Actually**: postpone; it heads towards myogenic failure.
- **Trap**: remove native kidneys routinely — **Actually**: keep unless indicated.
- **Trap**: routine stenting is evidence-based — **Actually**: no data; stent selectively.
- **Trap**: Deflux for graft VUR — **Actually**: poor resolution, higher obstruction risk.
- **Trap**: hydronephrosis grade reflects graft impairment — **Actually**: it does not correlate.
- **Trap**: isolated VUR explains a rising creatinine with hydronephrosis — **Actually**: suspect obstruction.